

Bike Rental Demand Analysis

Exploratory Data Analysis & Predictive Modeling Report

Dataset: Capital Bikeshare, Washington D.C. (2011–2012)

Daily records: 731 | Hourly records: 17,379

1. Introduction

Bike sharing systems represent a modern, automated approach to traditional bike rentals, where the entire process — from membership and rental to return — is handled without manual intervention. Users can rent a bike from one location and return it at a different one, making these systems highly convenient for short-distance urban travel. With over 500 bike-sharing programs operating worldwide, these systems play an increasingly important role in addressing traffic congestion, reducing environmental impact, and promoting public health.

This project uses the **Capital Bikeshare** dataset from Washington D.C., USA, covering rental activity across 2011 and 2012, available at two levels of granularity: daily aggregated counts (731 records) and hourly counts (17,379 records). Both datasets include rental counts alongside weather/seasonal information (temperature, humidity, windspeed, weather condition) and calendar details (season, holiday, working day).

Understanding what drives bike rental demand has real business value — operators need to predict demand accurately to ensure the right number of bikes are available at the right time and location, avoiding both shortages and idle inventory.

Objectives

- Perform a detailed exploratory analysis to uncover patterns and relationships between rental demand and environmental/seasonal factors.
- Build a predictive model that estimates daily rental count (*cnt*) from environmental and seasonal features.
- Compare multiple modeling approaches and recommend the best one for production use.
- Document challenges encountered and how they were resolved.

2. Data Understanding

The **day.csv** dataset contains 731 rows and 16 columns. Of these, 7 numerical columns (season, yr, mnth, holiday, weekday, workingday, weathersit) are categorical in nature — representing labels rather than continuous quantities. The true continuous columns are: temp, atemp, hum, windspeed, casual, registered, and cnt (the target variable). No missing values and no duplicate rows were found in the dataset.

Check	Result
Rows × Columns	731 × 16
Missing values	None
Duplicate rows	None
Categorical columns	season, yr, mnth, holiday, weekday, workingday, weathersit
Continuous columns	temp, atemp, hum, windspeed, casual, registered, cnt

3. Exploratory Data Analysis

3.1 Univariate Analysis — Distribution of Variables

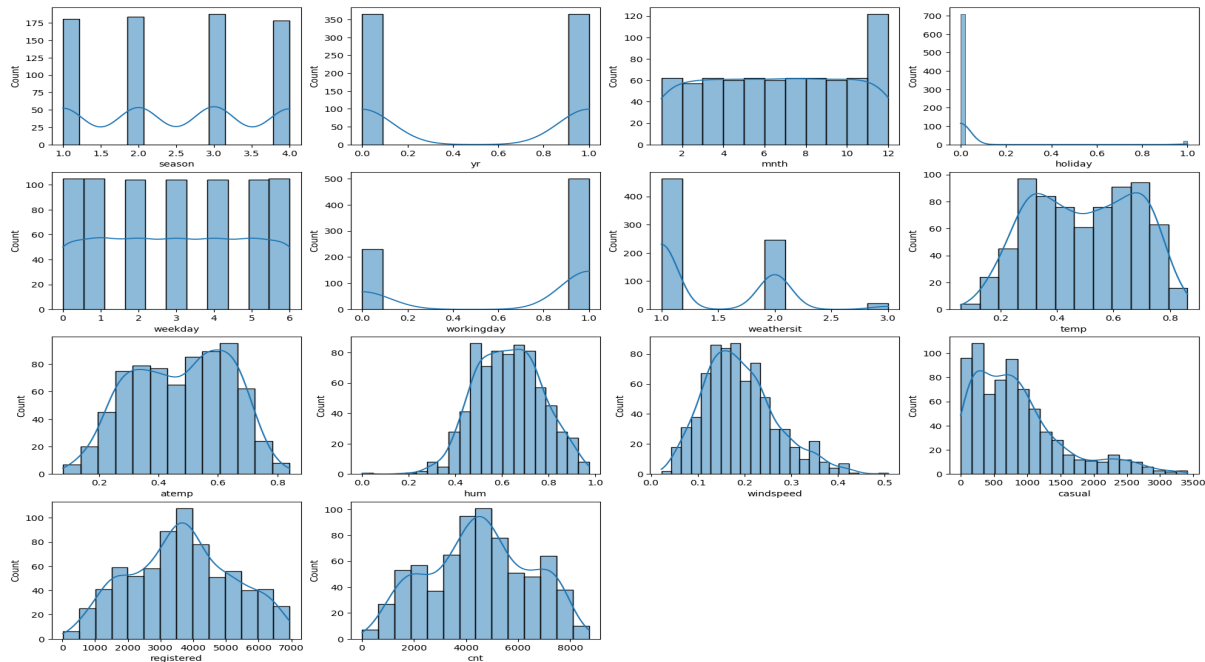


Figure 1: Distribution of all numerical/categorical variables in day.csv

- **season, yr, mnth, weekday:** Fairly evenly distributed — no category is heavily over/under-represented.
- **holiday:** Highly imbalanced — only ~2.9% of days (21 days) are holidays.
- **weathersit:** Highly imbalanced — majority of days (~460+) are Clear/Partly cloudy; extreme weather (category 3/4) is rare.
- **casual riders:** Strongly right-skewed — most days have few casual riders, with a long tail on favorable days.
- **registered riders:** More symmetric, suggesting consistent routine/commuting usage rather than weather-driven spikes.
- **cnt (target):** Slightly right-skewed, close to normal — most days between 3,000–6,000 rentals.

Key takeaway: holiday and weathersit are highly imbalanced, meaning the model will see few examples of holidays or extreme weather — this may limit prediction reliability for those specific conditions. Casual riders appear more sensitive to external factors (weather, weekends) than registered members.

3.2 Bivariate Analysis — Categorical Features vs. Rental Count

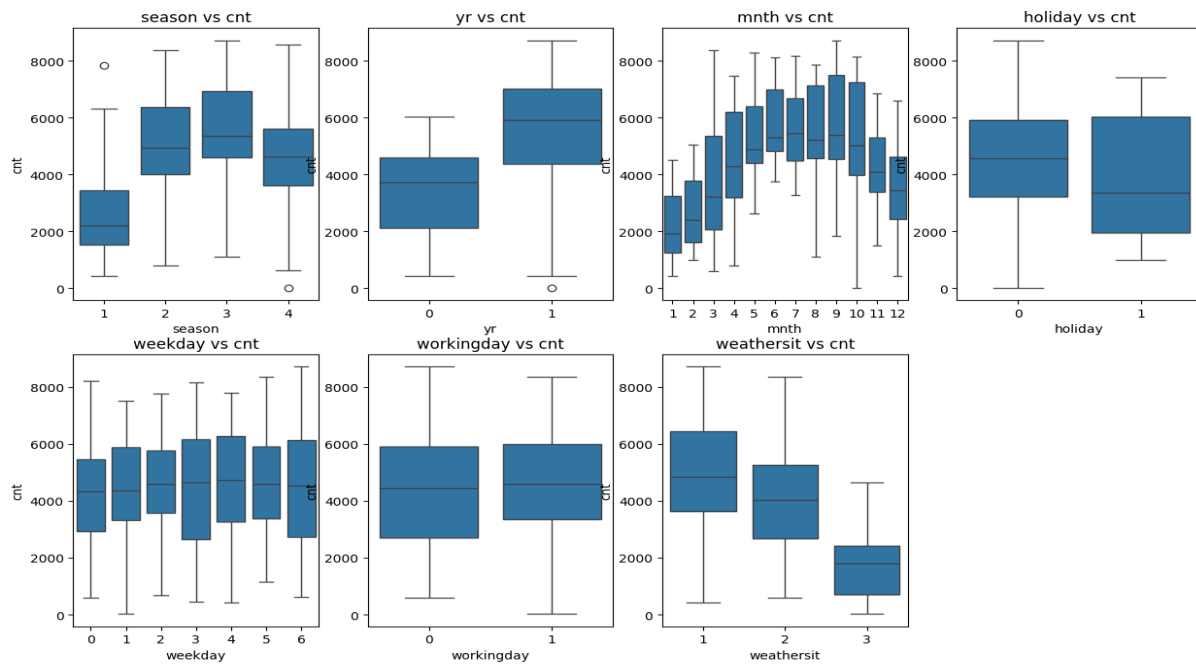


Figure 2: Rental count (cnt) distribution across categorical features

- **season:** Season 1 (winter) has the lowest median (~2,200); seasons 2–4 show higher medians (~4,900–5,400), with season 3 (fall) peaking.
- **yr:** 2012 shows a clearly higher median (~5,900) vs. 2011 (~3,700) — strong year-over-year growth.
- **mnth:** Clear seasonal rise from January, peaking June–September, declining toward December.
- **holiday:** Slightly lower demand on holidays (~3,500) vs. non-holidays (~4,500), though sample size is small.
- **weekday / workingday:** Minimal variation in total daily demand — day type alone is not a strong driver of the aggregate count.
- **weathersit:** Strong, clear pattern — Clear weather (~4,800 median) > Mist/Cloudy (~4,000) > Light Snow/Rain (~1,800).

Outlier Investigation

Two notable outlier patterns were investigated directly rather than removed blindly:

- **High-demand days (cnt > 8,000):** 12 days, all from 2012, mostly under clear weather — genuine high-demand days during a period of business growth, not data errors.
- **Extreme low-demand day (cnt < 100):** A single day — 29 October 2012, cnt = 22 — traced to **Hurricane Sandy** striking Washington D.C. This is a real-world event and was retained in the dataset rather than removed, to preserve realistic extreme-case behavior for the model.

3.3 Continuous Features vs. Rental Count



Figure 3: Scatter plots of continuous weather variables against rental count

- **temp / atemp:** Strong positive but non-linear relationship — demand peaks around $\text{temp} \approx 0.6\text{--}0.7$ and slightly drops at extremes.
- **hum (humidity):** Weak relationship, mostly scattered; slight drop only at very high humidity (>0.9).
- **windspeed:** Weak negative relationship — high wind slightly reduces rentals.

Key takeaway: temp/atemp are the strongest continuous predictors; the non-linear shape suggests tree-based models may outperform simple linear regression.

3.4 Correlation Analysis

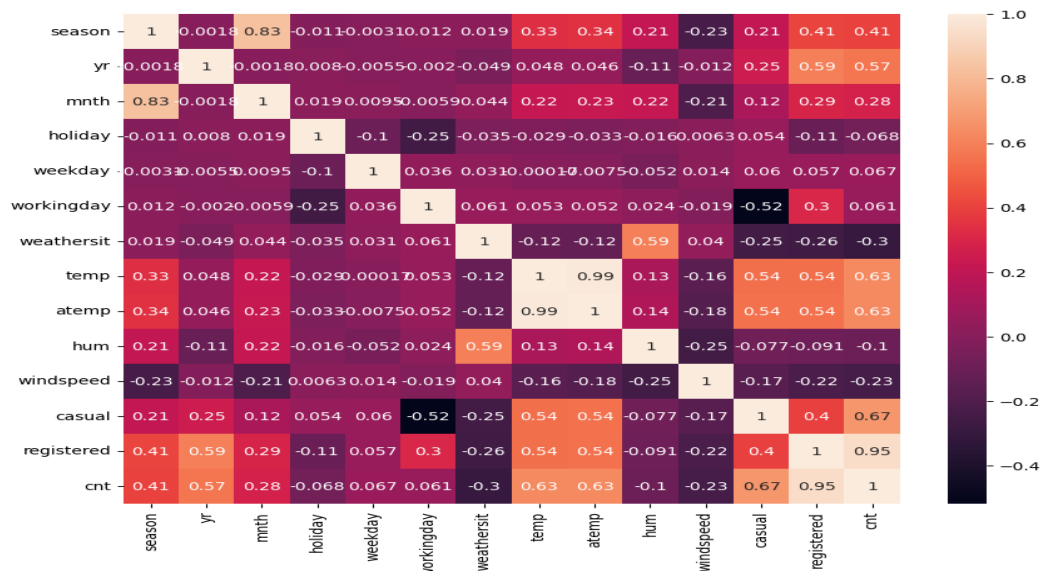


Figure 4: Correlation heatmap of all numeric features

- registered (0.95) and casual (0.67) correlate strongly with cnt, but this is a definitional artifact (cnt = casual + registered) and these were excluded from modeling to avoid data leakage.
- Among genuine predictors: temp/atemp (0.63), yr (0.57), season (0.41), weathersit (−0.30).
- weekday, workingday, and holiday show negligible correlation with total cnt (all below 0.07).
- **Multicollinearity detected:** temp & atemp (0.99 correlated) and season & mnth (0.83 correlated) — only one from each pair is needed.

3.5 Time Series Trend

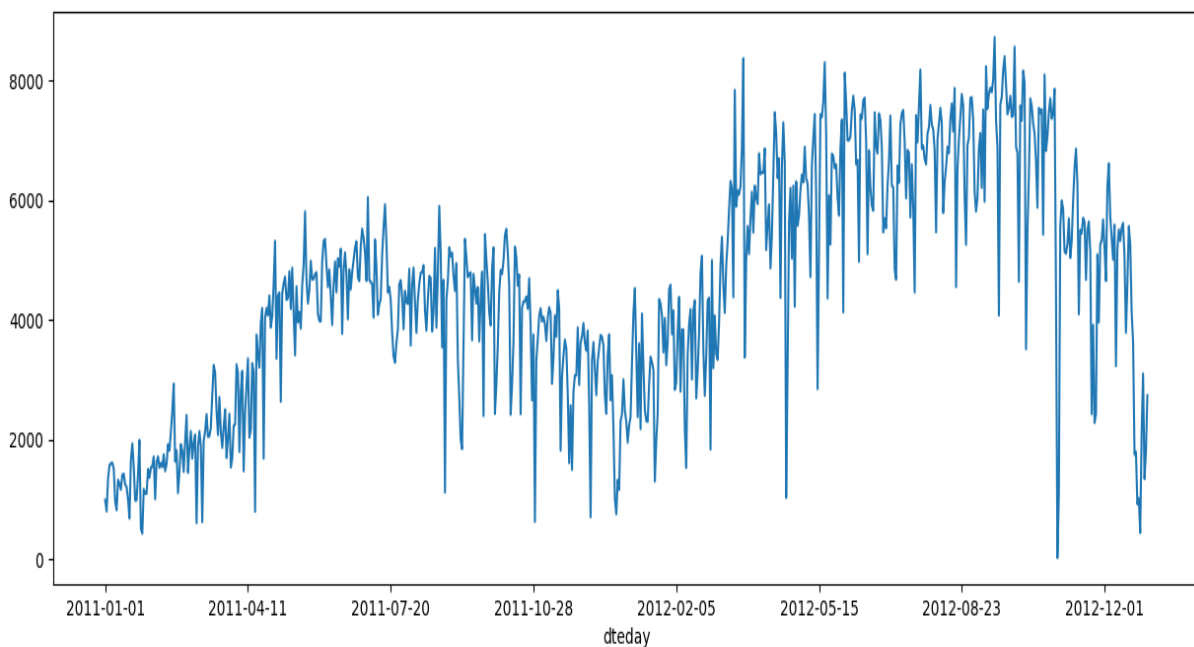


Figure 5: Daily rental count over time (2011–2012)

- Clear yearly seasonality: low in winter, peaking in summer/early fall, repeating across both years.
- Visible year-over-year growth: 2012 sits noticeably higher than the corresponding 2011 cycle.
- Sharp anomaly visible in late October 2012 — the Hurricane Sandy event identified earlier.

3.6 Hourly Demand Patterns (hour.csv)

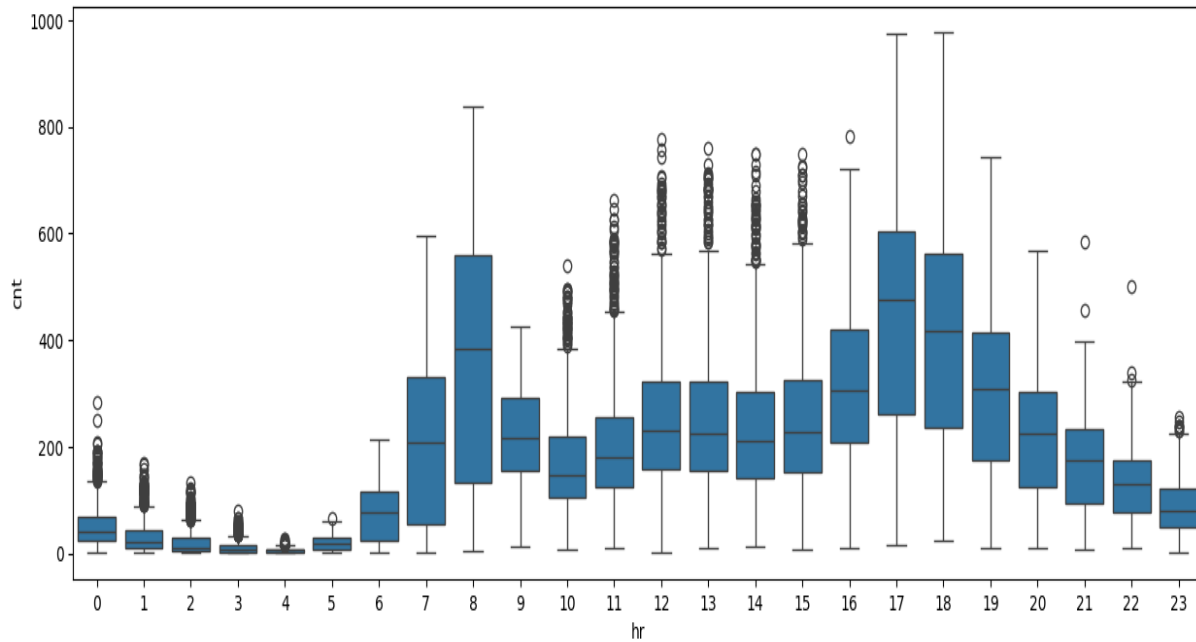


Figure 6: Rental count by hour of day

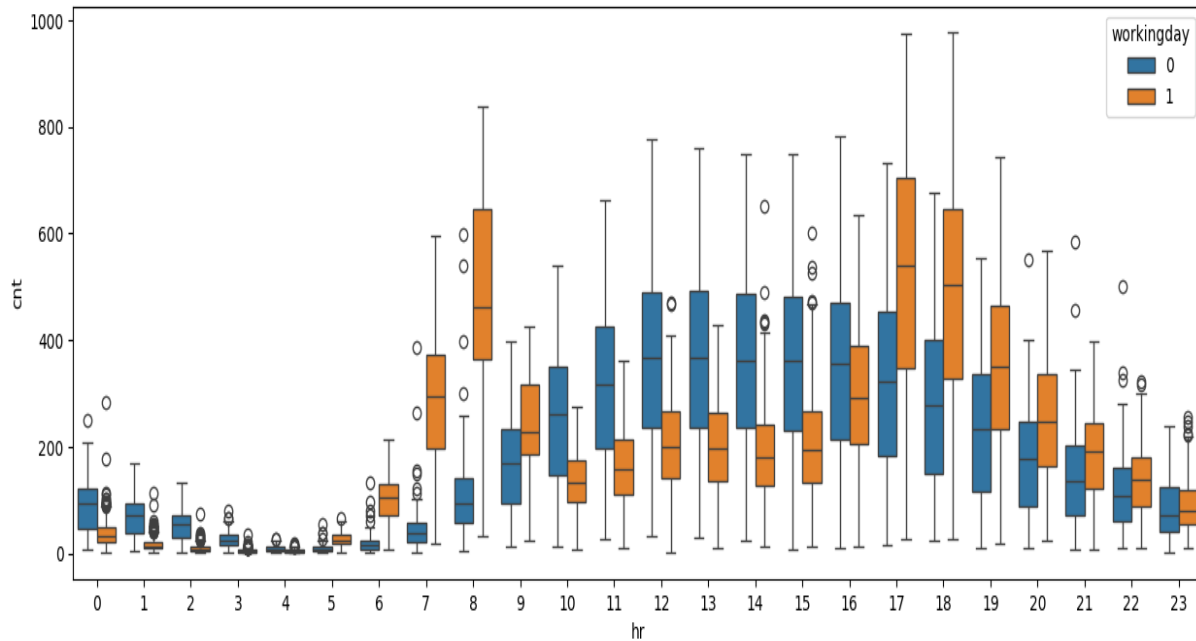


Figure 7: Rental count by hour, split by working day vs. non-working day

- **Overall:** Two distinct peaks — ~8 AM (morning commute) and ~17–18 (evening commute, the day's highest).
- **Working days:** Sharp bimodal pattern matching a classic office commute — visible only in hourly data, not daily aggregates.
- **Non-working days:** A single broad midday peak (10 AM–5 PM), consistent with leisure riding rather than commuting.

Key takeaway: The hourly breakdown reveals commuting behavior completely hidden in daily-aggregated data — confirming that registered (commute-driven) and casual (leisure-driven) riders have fundamentally different usage patterns.

4. Predictive Modeling

Before modeling, *casual* and *registered* were dropped (data-leakage risk, since cnt is their exact sum), along with *instant* and *dteday* (identifier / already encoded via derived features). Categorical columns (season, mnth, weekday, weathersit) were one-hot encoded. Data was split 80/20 into train and test sets. Four regression models were trained and compared.

Model	Train R^2	Test R^2	MAE	RMSE
Linear Regression	≈0.85	0.842	583.02	796.46
Decision Tree (depth=5)	0.874	0.788	649.33	921.47
Random Forest (tuned)	0.966	0.867	473.91	729.05
Gradient Boosting	0.952	0.885	485.23	679.22

4.1 Linear Regression (Baseline)

Explains ~84.2% of variance in daily rental count — a solid baseline. Predictions differ from actual values by ~583 bikes on average (~13% relative to the mean cnt of ~4,504). Because temp's relationship with cnt is non-linear, Linear Regression is structurally limited in capturing that pattern.

4.2 Decision Tree Regressor

An unrestricted tree achieved a perfect training R^2 (1.0) but only ~0.75 on test data — a textbook case of **overfitting**. Tuning `max_depth` to 5 gave the best balance (Train R^2 0.874, Test R^2 0.788), but even tuned, a single tree underperformed the Linear Regression baseline, motivating a move to ensemble methods.

4.3 Random Forest Regressor

Random Forest substantially outperformed both prior models (Test R^2 0.867–0.868). A systematic GridSearchCV (216 combinations, 5-fold cross-validation) confirmed that minimal restriction was near-optimal for this dataset size (~731 rows) — more aggressive tuning did not meaningfully improve results. Final selected configuration: `max_depth=15`, `n_estimators=200`, `min_samples_leaf=2` (Test R^2 = 0.867, MAE = 473.91, RMSE = 729.05).

4.4 Gradient Boosting Regressor — Best Performing Model

Gradient Boosting achieved the **highest test R^2 (0.885)** and **lowest RMSE (679.22)** among all models tested, with the smallest train-test gap (0.952 vs. 0.885) relative to Random Forest — indicating better generalization despite being a more complex, sequential ensemble. Its error-correcting, sequential-tree approach captures the non-linear temp-cnt relationship and weather/season interaction effects more effectively than any other model tested.

Key takeaway: Gradient Boosting Regressor is recommended for production use: it explains 88.5% of variance in daily demand with the smallest large-magnitude prediction errors. Random Forest remains a

strong, more interpretable alternative if simplicity or faster retraining is preferred over the marginal accuracy gain.

5. Challenges Faced & How They Were Resolved

Conflicting documentation

The README and project docs gave conflicting season-column mappings and slightly different temperature normalization formulas. Rather than trusting either blindly, this was flagged as a discrepancy to validate empirically if precise labels were ever required.

Data leakage risk

cnt = casual + registered exactly (verified across all 731 rows). Both columns were excluded from the feature set to prevent the model from trivially reconstructing the target.

Imbalanced categorical features

holiday (~2.9% positive) and weathersit (category 3 only ~20 days) are highly imbalanced — noted as a limitation on the model's reliability for rare conditions.

Extreme outlier — Hurricane Sandy

A single record (29 Oct 2012, cnt=22) was traced to a real weather event rather than a data error, and was retained to preserve realistic extreme-case behavior.

Overfitting in Decision Tree

An unrestricted tree hit Train $R^2=1.0$ but Test $R^2\approx 0.75$. Resolved via max_depth tuning, though this motivated moving to ensemble methods entirely.

Tuning trade-offs in Random Forest

GridSearchCV confirmed that minimal restriction was already near-optimal — a useful lesson that more aggressive tuning doesn't always help, especially on smaller datasets.

Multicollinearity

temp/atemp (0.99 correlated) and season/mnth (0.83 correlated) were flagged — tree-based models handle this reasonably well, but it matters for any future linear-model or feature-importance work.

6. Conclusion

This project analyzed two years (2011–2012) of Capital Bikeshare rental data to understand the factors driving daily bike rental demand and to build a predictive model for rental count.

Key EDA Findings

- Bike rental demand shows strong seasonality (low in winter, peaking in summer/fall) and clear year-over-year growth.
- Weather condition and temperature are the strongest environmental drivers, with temperature showing a non-linear relationship.
- Hourly analysis revealed a distinct commuting pattern on working days (two peaks: ~8 AM and ~5–6 PM) versus a broader leisure-driven pattern on non-working days — an insight invisible in daily-aggregated data.
- An extreme low-demand day was identified and traced to Hurricane Sandy, and retained as a genuine real-world event.

Modeling Results

Four models were built and compared — Linear Regression, Decision Tree, Random Forest, and Gradient Boosting. **Gradient Boosting achieved the best overall performance** (Test $R^2=0.885$, RMSE=679.22) and is recommended for production use, with Random Forest as a strong, more interpretable alternative.

This project demonstrates that both environmental/seasonal factors and systematic, non-linear ensemble modeling techniques are essential for accurately forecasting daily bike rental demand.